

TAMBACOUNDA, Senegal

WMO: 616870

Lat: 13.737N

Lon: 13.653W

Elev: 161

StdP: 14.61

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	63.8	65.8	21.5	16.2	82.5	25.2	19.3	80.2	22.2	82.6	20.0	84.6	5.6	40	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	22.8	107.9	72.6	106.3	72.3	104.9	72.2	83.3	94.3	82.1	92.0	81.2	90.8	7.3	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.7	160.6	89.1	79.5	154.1	87.6	78.8	150.5	86.9	47.4	94.1	46.1	92.0	45.0	91.1	90.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.7	15.3	13.4	58.6	111.4	2.6	1.9	56.7	112.7	55.2	113.8	53.8	114.9	51.9	116.3		
			45.3	85.7	2.0	2.0	43.9	87.2	42.7	88.3	41.6	89.5	40.2	90.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	86.0	80.4	84.5	89.5	92.9	93.3	89.8	84.4	82.3	82.6	85.6	85.0	81.4		
	DBStd	5.15	3.56	3.79	3.17	2.44	2.40	3.76	3.02	2.40	2.33	2.47	2.57	3.17		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	13130	941	967	1225	1287	1341	1195	1068	1000	979	1105	1049	973		
	CDD65	7655	476	547	760	837	876	745	603	535	529	640	599	508		
	CDH74	103826	5614	7331	11554	13552	14257	11248	7467	5675	5683	7845	7573	6027		
	CDH80	61673	3135	4414	7506	9304	9826	7100	3574	2226	2446	4127	4548	3467		
(14) Wind	WSAvg	6.8	8.2	8.3	7.4	6.7	7.0	7.6	6.8	6.3	5.1	5.2	5.8	7.7		
(15) Precipitation	PrecAvg	30.4	0.0	0.0	0.0	0.0	0.7	3.7	6.6	8.8	7.4	2.4	0.0	0.0		
	PrecMax	49.4	0.0	0.9	0.2	1.7	4.2	8.5	13.2	18.7	17.4	8.1	0.5	0.3		
	PrecMin	17.3	0.0	0.0	0.0	0.0	0.0	0.3	0.2	1.7	2.2	0.0	0.0	0.0		
	PrecStd	7.8	0.0	0.1	0.0	0.2	1.0	1.9	2.5	3.9	3.5	1.9	0.1	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	100.8	105.3	108.0	110.0	110.7	107.3	98.4	93.9	95.2	101.8	103.3	102.0		
		MCWB	67.6	68.9	70.3	71.9	75.7	77.5	80.5	81.5	81.4	76.3	71.3	66.8		
	2%	DB	98.4	102.5	106.1	108.0	107.8	104.2	95.4	91.8	93.3	98.9	101.2	100.0		
		MCWB	66.3	68.5	69.6	72.0	74.0	76.8	79.5	80.2	81.2	77.9	71.4	67.1		
	5%	DB	96.4	100.5	104.4	106.7	106.0	102.0	93.6	90.3	91.7	97.0	99.9	97.8		
		MCWB	65.5	67.7	69.5	71.9	74.0	77.0	79.2	79.6	80.6	78.8	71.1	66.7		
	10%	DB	93.9	98.4	102.5	104.9	104.1	99.7	91.8	89.2	90.2	95.1	98.2	95.2		
		MCWB	64.3	66.5	69.0	71.4	74.2	76.6	78.9	79.3	80.1	79.2	71.0	65.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.5	75.5	76.5	79.8	83.1	85.5	84.6	83.7	83.6	84.1	81.7	74.3		
		MCDB	94.1	98.0	100.9	103.1	104.7	101.4	93.8	91.7	91.2	94.2	96.3	93.5		
	2%	WB	70.3	72.4	74.5	77.2	80.1	81.9	82.3	81.8	82.4	82.4	78.9	71.1		
		MCDB	93.0	96.2	98.1	101.5	100.2	96.3	90.7	88.8	90.4	93.2	92.2	93.0		
	5%	WB	67.6	70.1	72.8	75.5	78.2	80.0	80.8	80.6	81.4	81.5	77.4	69.3		
		MCDB	91.4	95.2	96.5	99.5	97.8	93.8	89.7	87.8	89.3	91.9	89.6	91.0		
	10%	WB	65.5	67.8	71.0	74.0	76.9	78.9	79.8	79.8	80.6	80.7	75.8	67.5		
		MCDB	90.6	94.3	96.0	97.6	96.5	92.4	88.7	86.8	88.4	90.8	87.2	90.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	27.0	26.9	26.2	25.3	22.8	19.1	14.8	13.7	15.6	20.2	27.8	27.6		
		MCDBR	30.5	29.5	28.5	27.1	25.4	21.8	16.6	15.3	17.4	23.1	30.0	31.4		
	5% WB	MCWBR	13.7	13.0	11.6	9.7	8.3	7.0	6.0	6.6	7.9	7.2	9.7	12.2		
		MCWBR	28.1	27.5	26.1	24.6	22.2	19.5	15.6	14.7	16.7	19.7	24.8	28.4		
(40) Clear-Sky Solar Irradiance	taub		0.517	0.564	0.671	0.670	0.701	0.706	0.622	0.562	0.561	0.597	0.531	0.488		
		taud	1.912	1.800	1.587	1.636	1.596	1.653	1.847	1.999	1.977	1.878	1.957	2.026		
	Edn at Noon		247	241	220	220	209	206	225	242	243	230	241	252		
		Edh at Noon	60	70	88	84	86	80	66	57	59	63	57	52		
(44) All-Sky Solar Radiation	RadAvg		1686	1877	2097	2205	2137	1967	1771	1684	1729	1780	1698	1620		
	RadStd		93	84	76	74	84	65	69	68	80	47	71	67		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page